PS1-2

On-going task development to measure behavioral outcomes of pattern completion in the conceptual domain in 6–8-year-old children

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Theoretical background

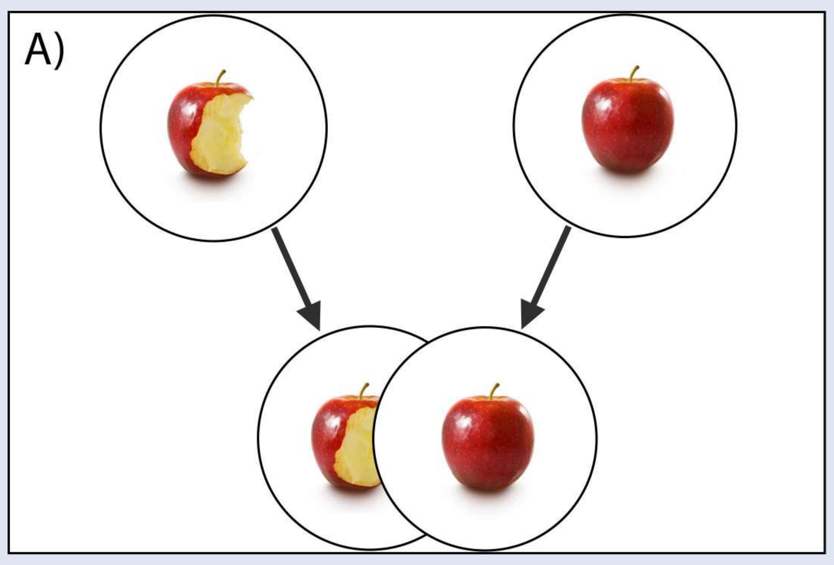
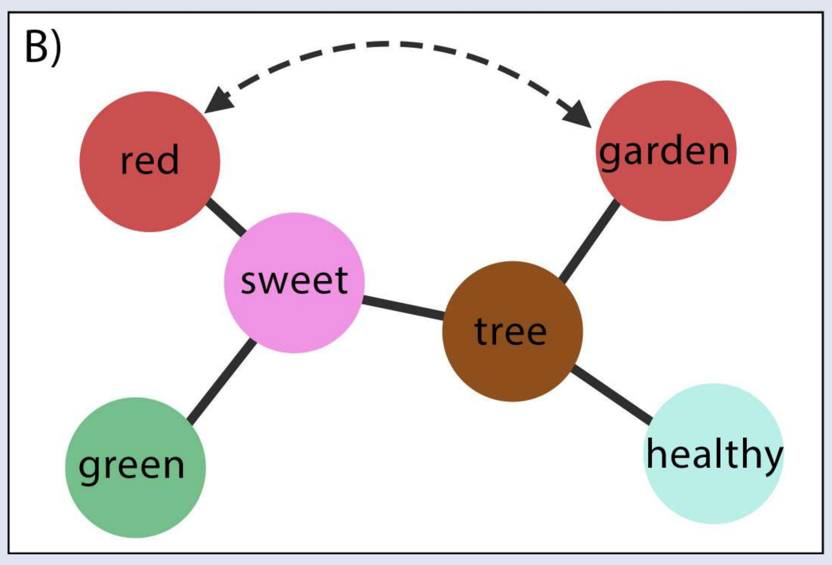


Figure 1. A) Theoretical model of hippocampal pattern completion: a partial cue of an already encoded representation is completed as the original representation. B) Theoretical model of pattern completion within a concept space: the hippocampus may bind together otherwise unrelated features of the same concept.



Hippocampal pattern completion is a neural operation that allows retrieval of a representation based on a partial or degraded input [1].

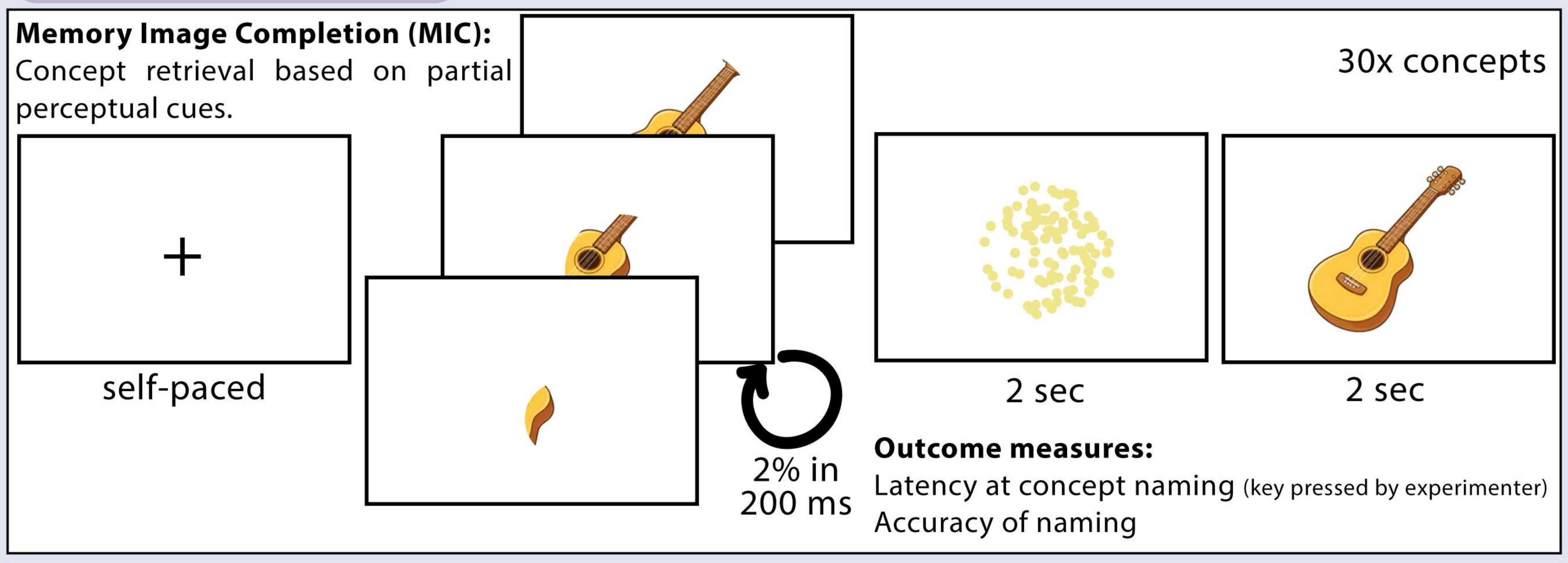
Recent models suggest that the hippocampus might employ pattern completion to dynamically connect otherwise unrelated representations of features shared by the same concept [2].

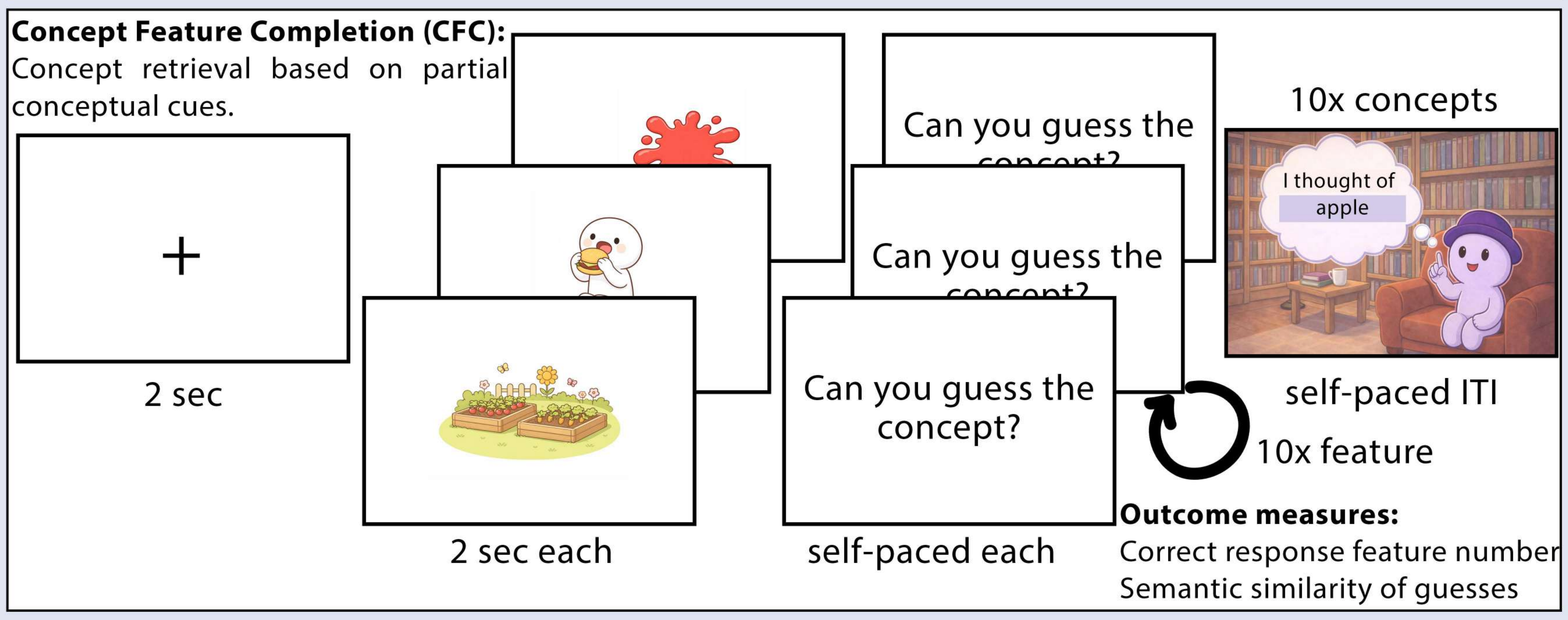
Such a framework assumes that partial conceptual and perceptual information rely on the same neural operation during retrieval of the target representation. This shared neural operation should therefore account for a behavioural link between concept retrieval via partial perceptual and conceptual cues.

H1: We expect a positive correlation between the amount of information neccessary for concept retrieval based on visual and semantic information.

H2: We expect a positive correlation between accuracy during concept retrieval based on visual and conceptual information.

Task design





Stimuli

In an on-going concept norming study, we are collecting feature descriptions for 300 concepts (target = 20 response per concept) from 6-to-8 years old Hungarian children (n = 518, M_{age} = 7,76 years, SD_{age} = 1,24 years, 269 females). We extracted 10 features with frequency > 1 for each concept, and created visual representations via ChatGPT [3].

A)

Apples are **red**, and are **circle shaped**...

Apples **can be found in gardens** and **grow on trees**...

Apples are very **delicious**...

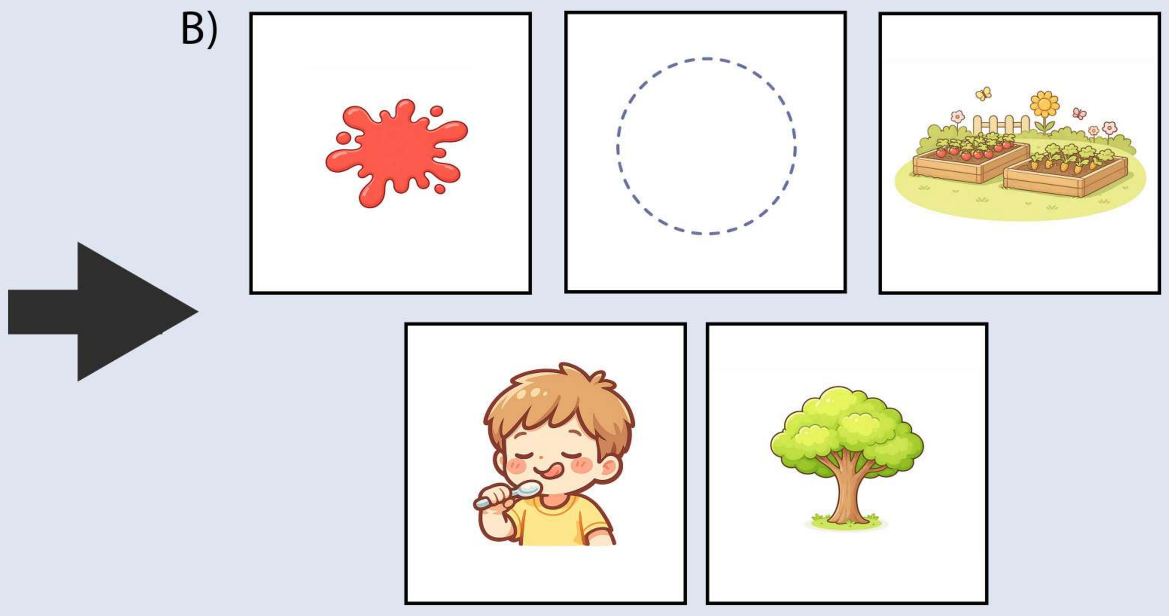


Figure 2. A) Sample transcriptions of features produced by children during the norming study collecting feature description.. B) Visual representations of concept features were shared across concepts, created via generative AI.

Participants

We recruited **14 Hungarian children** (M_{age} = 6.5 years, SD_{age} = 0.85 years, 7 females) who completed **CFC first, followed by MIC** in a single session. For the **analyses involving CFC**, we limited the sample to **eight children** (M_{age} = 6.56 years, SD_{age} = 0.88 years, 4 females), because we substantially changed the task design after the first five participants produced floor effects on accuracy, and excluded an outlier at 3SD.

Exploratory results

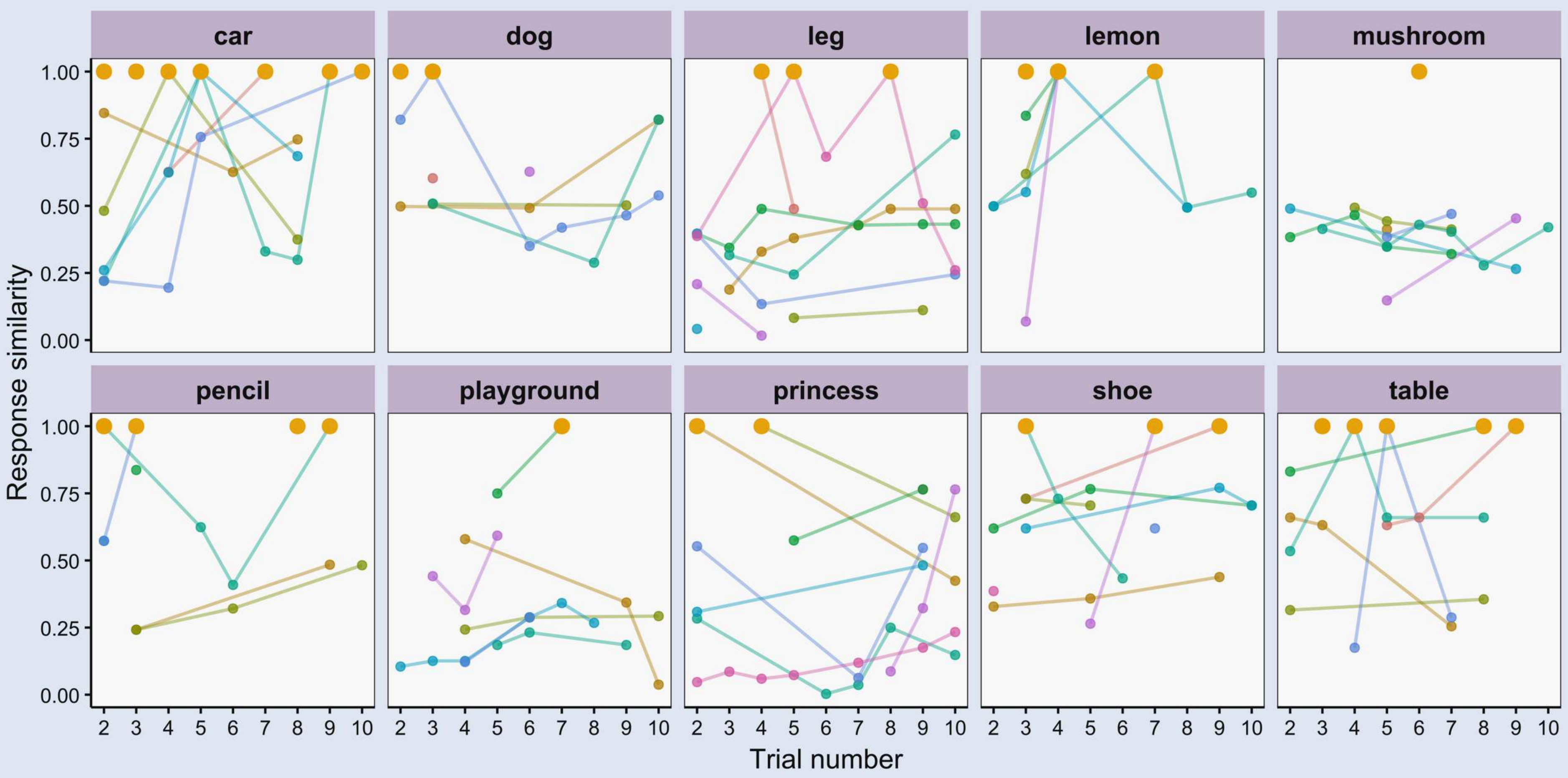


Figure 3. Each facet shows one of the 10 concepts in the CFC . The x axis shows the order of presented features, starting from the second feature because no responses was prompted after the first feature to facilatete integration. The y axis shows cosine similarity between word-vectors from a Hungarian word-embedding [4,5] comparing each response and the target concept. Dots show each response coloured by participant (in amber if it matched the target concept exactly).

Figure 4. Facets show each of the 10 concepts in the CFC . Responses are plotted as text with text size indicating response frequency and colour indicating cosine similarity to the target concept ranging from 0 (in blue) to 1 (exact match, in orange).

Main results

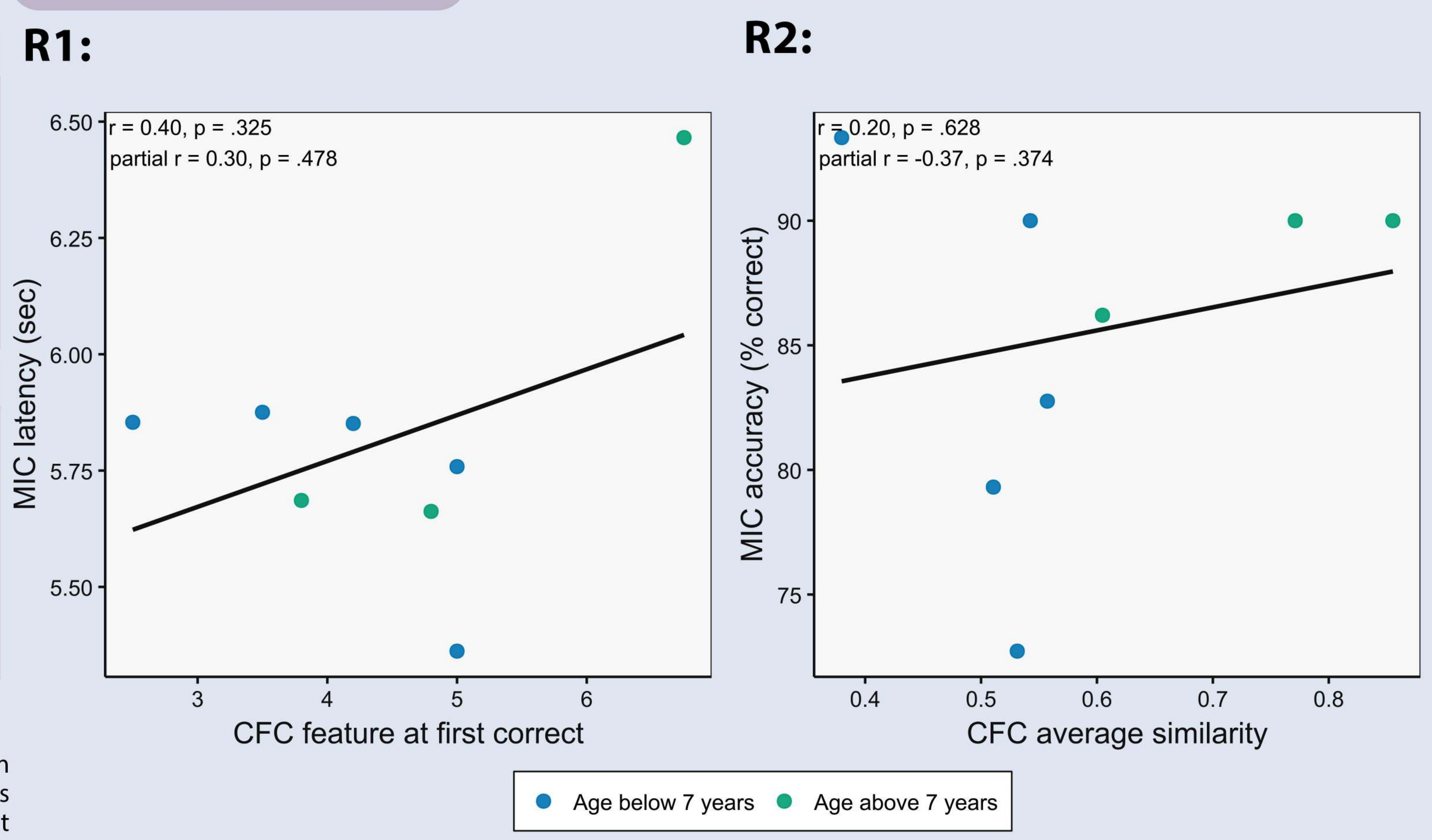


Figure 6. Facets show correlations with and without partialing out age in months between MIC and CFC task outcomes. Left: number of features presented before a correct response in the CFC task correlated with reaction time to correct response in the MIC task. Bottom: average cosine similarity between CFC responses and target concepts correlated with percent-correct accuracy in the MIC task. Dots represent participant-specific averages, coloured by age (blue for participants below age seven, green for participants above age seven).

Discussion

On-going pilots probing concept retrieval by partial perceptual or conceptual information hint at a shared behavioural process. However, results remain tentative due to the small sample size and age-related confounds in this pilot dataset. Preliminary results indicate that the amount of perceptual information — i.e., size of partial concept image shown — and the amount of conceptual information — i.e., number of concept-related features presented — neccessary for correct response may be linked, partly independent of age effects.

Outstanding questions:

- How can we select target concepts for CFC that produce sufficient between-subject variability while introducing no floor effects for the sample?
- How can syntactic meaning be conveyed through images depicting conceptual features?

Acknowledgments

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